

# MARYAM REZAAEE

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A computer science researcher specialising in LLM Interpretability and Cognitively-Inspired AI. My objective is to develop robust, explainable systems by investigating the integration of human inductive biases and symbolic reasoning structures into neural models. My foundational experience includes a thesis on LLM input-output interactions and research in concept-based interpretability. I am now expanding this focus towards neurosymbolic architectures, evidenced by projects in program generation. I seek to advance this research with the aim of developing models capable of more robust and human-aligned reasoning.

## Education

### M.Sc. in Computer Science

Sep 2023 – Present

Department of Mathematical Sciences, Sharif University of Technology, Iran AVG: 18.57 / 20.00

Thesis: *Interpretability in Generative Models: Investigating the Mechanisms Behind Output Generation in Large Language Models*

### B.Sc. in Computer Science

Sep 2019 – Sep 2023

Department of Mathematical Sciences, University of Isfahan, Iran AVG: 16.66 / 20.00

Project: *Understanding the Application of Machine Learning in Cell Signalling Pathway Analysis*

## Research Experience

### Exploring Neurosymbolic AI and Cognitive Architectures

Jul 2025 – Present

Independent Researcher

*Sharif University of Technology, Tehran, Iran*

- Conducting a self-directed literature review on the integration of neural networks with symbolic reasoning systems to inform future research.
- Studying key topics including program synthesis, logic-based model integration, and the role of human inductive biases in cognitive architectures.
- Implemented a seq2seq model for program generation as a practical exploration of neurosymbolic principles.

### Concept-Based Interpretability for Retrieval-Augmented Generation (RAG) Systems on Large Language Models

May 2025 – Present

Research Mentor under the supervision of Dr. Fatemeh Seyyedsalehi

*Trustworthy and Generative Machine Learning (TGML) Lab, Sharif University of Technology, Tehran, Iran*

- Mentoring a B.Sc. student in developing a novel framework for RAG interpretability.
- Investigating the application of Concept Bottleneck Models (CBMs) to explain the internal generation processes.
- Developing methods using Concept Activation Vectors (CAV) and Automatic Concept-based Explanations (ACE) to enhance the transparency and traceability of RAG outputs.

## A Concept Level Energy-Based Framework for Interpreting Black-Box Large Language Model Responses

[Link](#) Feb 2025 – Present

Graduate Researcher under the supervision of Dr. Fatemeh Seyyedsalehi

TMML Lab, Sharif University of Technology, Tehran, Iran

- Proposed a model-agnostic, post-hoc framework (ESCI) to interpret black-box LLMs accessed only via APIs.
- Designed an energy model to quantify the conceptual links between user prompts and LLM-generated responses.
- Trained an efficient interpreter network to provide sentence-level importance scores.
- Established proof-of-concept results and currently conducting extensive experimental validation to refine the methodology for submission to ACL; for preliminary preprint, view the above link.

## Interpretability in Generative Models: Investigating the Mechanisms Behind Output Generation in Large Language Models

Sep 2024 – Present

Master's Thesis Researcher under the supervision of Dr. Fatemeh Seyyedsalehi

TMML Lab, Sharif University of Technology, Tehran, Iran

- Conducted a comprehensive review of intrinsic, post-hoc, concept-based, and mechanistic interpretability methods for generative models, with a primary focus on LLMs.
- Developed a post-hoc framework to interpret instance-based input-output attribution in LLM generation.
- Analysing internal model mechanisms and their influence on output generation.
- Investigating ways to leverage interpretability findings to improve model robustness, alignment, and performance.

## Understanding the Application of Machine Learning in Cell Signalling Pathway Analysis and Developing a DEG Analysis Handbook

[Link](#) Feb 2023 – Sep 2023

Bachelor's Project Researcher under the supervision of Dr. Fatemeh Mansoori

University of Isfahan, Isfahan, Iran

- Conducted a literature review on the application of ML and DL (e.g., SVMs) in bioinformatics.
- Analysed and documented methodologies for Gene Expression and Differentially Expressed Gene (DEG) analysis.
- Authored a technical handbook and implementation roadmap for future students to use in DEG analysis.

## Foundational Research in Cognitive Neuroscience

Aug 2016 – Feb 2017

Team Researcher (1st Kashan Neuroscience Workshop for Students)

Kashan University of Medical Sciences, Kashan, Iran

- Achieved 2nd Place in a multi-stage research competition for a portfolio on neuroscience and psychology.
- Researched and presented on cognitive topics, including the neural basis of creativity (divergent thinking), hypnotic analgesia, and psychological models of learning/cognitive styles.
- Conducted foundational reviews on core neuroanatomy (Limbic System), the brain-mind problem, and applied clinical neuroscience as part of the workshop curriculum.

## Three Centuries in Pursuit of the Theory of Everything

Oct 2015 – Feb 2016

Independent Researcher (Khawrazmi Youth Award)

Farzanegan School (NODET: National Organization for Development of Exceptional Talents), Kashan, Iran

- Authored a comprehensive research paper on the historical and theoretical development of a unified field theory in physics, analysing key milestones from classical mechanics to modern physics.
- Awarded 2nd Place in the Physics category at the Khawrazmi Youth Award (Kashan regional).

## Teaching Experience

### Relevant to Academic Major

- **Head Teaching Assistant: *Generative Models* (M.Sc.)** Sep 2025 – Present  
Dr. Fatemeh Seyyedsalehi, Sharif University of Technology
- **Teaching Assistant: *Machine Learning Operations* (B.Sc.)** Sep 2025 – Present  
Dr. Fatemeh Seyyedsalehi, Sharif University of Technology
- **Head Teaching Assistant: *Machine Learning* (B.Sc.)** Sep 2025 – Present  
Dr. Mohsen Alambardar, University of Isfahan
- **Instructor: *Fundamentals of Technological Product Development*** Jul 2024 – Sep 2024  
Tehran University of Medical Sciences
- **Head Teaching Assistant: *Advanced Programming* (B.Sc. Math & CS)** Jan 2023 – Jun 2023  
Dr. Fatemeh Mansoori, University of Isfahan
- **Head Teaching Assistant: *Operating Systems* (B.Sc.)** Jan 2023 – Jun 2023  
Dr. Mojtaba Rafiee, University of Isfahan
- **Instructor: *Algorithms and Data Structures* (B.Sc.)** Nov 2022 – Feb 2023  
Private Classes, University of Isfahan
- **Instructor: *Python Programming* (Elementary to Advanced)** Sep 2019 – Feb 2023  
Private Classes, University of Isfahan

### Additional Topics

- **Instructor and Mentor: *Graphic Design with Adobe Illustrator*** Jan 2023 – Jun 2023  
Academic Association of Mathematics and Computer Science (AMCSUI), University of Isfahan
- **Instructor: *English Language* (Intermediate to Advanced)** Sep 2016 – Nov 2023  
Private Classes

## Selected Course Projects

- **System 2 AI** (M.Sc.) Feb 2025 – Jun 2025  
Dr. M.H. Rohban & Dr. M. Soleymani, Sharif University of Technology  
Audit  
PY Program Generation w/ Seq2Seq Models PY Symbolic Regression PY GraphRAG PY LLM Agent
- **Deep Learning** (M.Sc.) Feb 2025 – Jun 2025  
Dr. Fatemeh Seyyedsalehi, Sharif University of Technology  
Mark: 20.00 / 20.00  
PY GPT2 PY Math Reasoning in LLMs PY PINN PY Atari DQN PY SSL & Contrast. Learning PY FNN from Scratch

**Generative Models (M.Sc.)**

Sep 2024 – Feb 2025

Dr. Fatemeh Seyyedsalehi, Sharif University of Technology

Mark: 18.00 / 20.00

PY RNN vs Transformer PY Diffusion/DDPM PY EBM PY  $\beta$ -VAE PY Pix2Pix PY Norm. Flows PY Bayesian Network

**Machine Learning (M.Sc.)**

Sep 2024 – Feb 2025

Dr. Ali Sharifi Zarchi, Sharif University of Technology

Mark: 20.00 / 20.00

PY Neural Decompo. PY LoRA from Scratch PY Word2Vec PY Knowl. Distil. (KD) & CLIP PY MobileNet, Eff. & KD.

**Matrix Computations (M.Sc.)**

Sep 2023 – Feb 2024

Dr. Mohammad Reza Razvan, Sharif University of Technology

Mark: 19.00 / 20.00

PY SVD RGB-Image Compressor from Scratch MATLAB Comparing Linear Solvers Implemented from Scratch

**Computer Networking (B.Sc.)**

Feb 2023 – Jun 2023

Dr. Mohsen Alambardar, University of Isfahan

Mark: 18.75 / 20.00

PY Group Chat Application with TCP Protocol and Highspeed Data Sharing

**Artificial Intelligence (B.Sc.)**

Sep 2022 – Feb 2023

Dr. Fatemeh Mansoori, University of Isfahan

Mark: 19.50 / 20.00

PY AI Search Algorithms for Solving Mazes PY MDP Gridworld Solver with Value & Policy Iteration

**Cryptography (B.Sc.)**

Sep 2022 – Feb 2023

Dr. Mojtaba Rafiee, University of Isfahan

Mark: 19.50 / 20.00

Feasibility of Proving Authenticity and Data Integrity for P2P Using Hash

**Fundamentals of Operating Systems (B.Sc.)**

Feb 2022 – Jun 2022

Dr. Mojtaba Rafiee, University of Isfahan

Mark: 19.50 / 20.00

PY Reader-Writer Lock Implementation with Three Priorities

**Mathematical Databases (B.Sc.)**

Feb 2022 – Jun 2022

Dr. Fatemeh Mansoori, University of Isfahan

Mark: 19.60 / 20.00

PostgreSQL University Course Registration System PostgreSQL Academic Transcript Generator & Analyzer

**Mathematical Software (B.Sc.)**

Feb 2022 – Jun 2022

Dr. Najmeh Hoseini, University of Isfahan

Mark: 17.75 / 20.00

MATLAB Audio-Signal Analysis MATLAB CLI University Portal for Grade Management

**Algorithms & Data Structures (B.Sc.)**

Feb 2022 – Jun 2022

Dr. Jaafar Almasizadeh, University of Isfahan

Mark: 20.00 / 20.00

PY Chinese Postman PY Huffman Compressor PY Max-Flow Altered PY Prime Sieves Anal. PY Topol. Sort Anal.

**Advanced Programming (B.Sc.)**

Feb 2020 – Jun 2020

Dr. Jaafar Almasizadeh, University of Isfahan

Mark: 20.00 / 20.00

PY RSA Cryptosystem Implementation from Scratch PY CLI Banking System

## Selected Talks

- ReAGent: A Model-Agnostic Feature Attribution Method for LMs** [Link](#) Aug 2025  
 Trustworthy & Generative ML (TGML) Lab, Sharif University of Technology
- Interpreting Language Models with Contrastive Explanations** [Link](#) Jun 2025  
 TGML Lab, Sharif University of Technology
- Propagating Universal Perturbations to Attack LLM Guard-Rails** [Link](#) May 2025  
 Deep Learning Seminars, Sharif University of Technology
- Universal and Transferable Adversarial Attacks on Aligned LMs** [Link](#) May 2025  
 Deep Learning Seminars, Sharif University of Technology
- An Exploration of Concept-Based Interpretability Methods** [Link](#) Mar 2025  
 TGML Lab, Sharif University of Technology
- DiT: Scalable Diffusion Models with Transformers** [Link](#) Feb 2025  
 Generative Models Seminars, Sharif University of Technology
- Saliency-Guided Training: Interpretability and Robustness in DNNs** [Link](#) Jan 2025  
 Machine Learning Seminars, Sharif University of Technology
- Model Sparsity Can Simplify Machine Unlearning** [Link](#) Dec 2024  
 Machine Learning Seminars, Sharif University of Technology
- A Foray into Bioinformatics: ML in Cell Signalling Pathway Analysis** [Link](#) Jun 2023  
 Bachelor's Project Presentation, University of Isfahan
- Key Takeaways from the 19th Iranian Conference on HPE** May 2018  
 Invited Speaker, Post-Conference Seminar, Kashan University of Medical Sciences

## Technical Skills

### Areas of Expertise

ML/DL



GMs/LLMs



XAI



NeSy AI



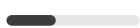
Reasoning



RAG



MLOps



Bioinform



CogNeuro

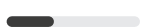


### Languages / Core Tools

Python



R



MATLAB



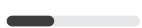
PostgreSQL



LaTeX



JavaScript



HTML/CSS



Sockets



Git/Bash



### Modules / Frameworks

PyTorch



TensorFlow



Scikit-learn



NumPy



Pandas



Transformers



Diffusers



Threading



RISC-V



### Technical Software

Illustrator



InDesign



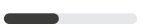
Photoshop



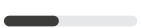
Premiere Pro



After Effects



Audition



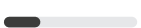
Figma



Canvas



Docker



## Leadership & Project Management

- **Conceptual Prototype Designer** Jul 2024 – Aug 2024  
"Siba," the Interprofessional Gamified Platform for Intelligent Education and Evaluation of Medical, Nursing, and Pharmacy Students | *16th Scientific Olympiad for Medical Science Students of Iran*
- **Founder, Manager, Lead Designer** May 2023 – Mar 2024  
"PsyCity," the National Competitive and Educational Game for Students of Computer Science
- **Committee Representative** Apr 2023 – May 2024  
Assembly for the Academic Associations | *University of Isfahan*
- **Association President, Head of Content Creation Branch** Oct 2022 – May 2024  
Association for Mathematics & Computer Science (AMCSUI) | *University of Isfahan*
- **Content Lead** Mar 2022 – Apr 2022  
"Varpharin," the Gamified Event for Diabetic Patient Management with an Interprofessional Approach | *AMEE Grant Recipient, Presented at AMEE 2023 (Short Communication)*

## Professional Activities & Service

- **Invited AI Panelist** Aug 2025  
Symposia on AI in Health Professions Education | *International AMEE 2025 Conference, Live*
- **English Interpreter & Student Task Force** May 2023  
International Guests & Presenters | *24th Iranian Conference on Health Professions Education*
- **UI/UX & Graphic Designer** Since 2018  
Academic & Institute Projects | *Freelance*
- **Academic & Technical Translator** Since 2016  
Book & Article Publication | *Freelance*

## Awards

|                                                                                                            |                                                                                                 |                                                                                                 |
|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Honouree: <b>Tech &amp; Innovation</b> 2024<br><i>17th International Harekat Festival, Iran</i>            | Top: <b>Digital Content</b> 2024<br><i>17th International Harekat Festival, Iran</i>            | Top: <b>Best Science Assoc.</b> 2024<br><i>17th University-Level Harekat Festival, UI, Iran</i> |
| Top: <b>Best &amp; Most Creative Pres.</b> 2024<br><i>17th University-Level Harekat Festival, UI, Iran</i> | 1st: <b>Most Active Association</b> 2024<br><i>Tadaee Festival, University of Isfahan, Iran</i> | Honouree: <b>Tech &amp; Innovation</b> 2024<br><i>Roshana Festival, Science Ministry, Iran</i>  |
| Honouree: <b>Educational Assoc.</b> 2023<br><i>16th International Harekat Festival, Iran</i>               | Honouree: <b>Best Team</b> 2018<br><i>Avicenna Student Creativity Cup, Iran</i>                 | 2nd: <b>Best Team</b> 2016<br><i>1st Kashan Neuroscience Workshop, Iran</i>                     |
| 2nd: <b>Best Research in Physics</b> 2016<br><i>18th Khawrazmi Youth Award, Kashan, Iran</i>               | 1st: <b>Poetry</b> 2016, 2017, 2018<br><i>Regional Cultural-Artistic Cup, Kashan, Iran</i>      | 1st: <b>Short Story</b> 2016, 2017<br><i>Regional Cultural-Artistic Cup, Kashan, Iran</i>       |

## Languages

● — Full Proficiency in **English**

● — Native Fluency in **Persian (Farsi)**